Tool Effector eXchange Array System (TEXAS)

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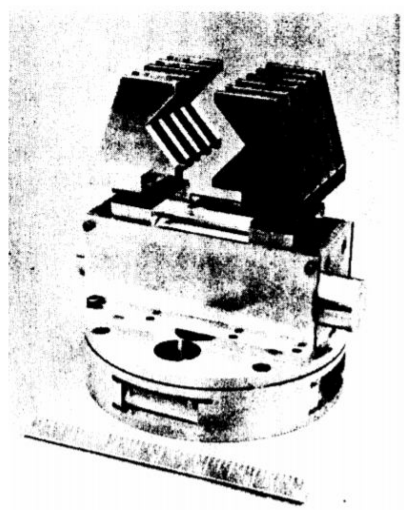
7.2.2

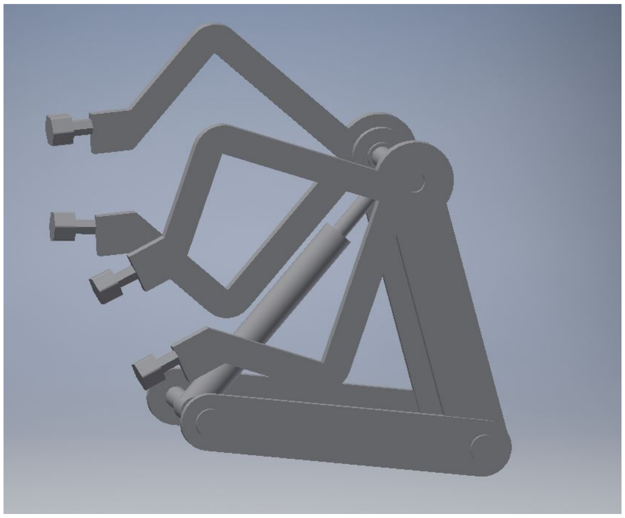
Main Body:

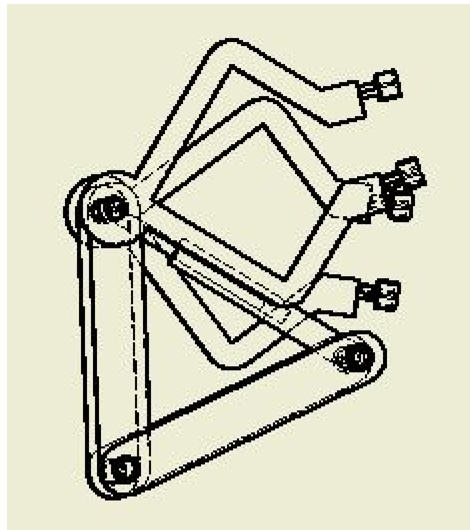
Manipulators: for the manipulator section, we will be focusing on the reduction of components as well as the weight of the gripper (end effector). The manipulator itself will not be affected. End effectors is the mechanism with one or more degrees of freedom, its very important as it physically grasps and manipulates objects in the work environment. It can be helpful for inspections, servicing and repair of space stations, as well as satellites, and it requires a simple manufacturing for its parts. For a gripper mechanism, the OMV Smart Hand is the best choice, one of the main advantages are that it has a simple design (useful for 3-D printing), its reliable, it has a relatively high clamping force strength, smart sensing and it offers a relatively light end effector. The Smart Hand (single DOF effector) is an advanced gripper with a variety of sensors in and around the hand, which helps with the object recognition and grasping, it can be used for square or cylindrical objects. The maximum opening of its tips is of 6.5 cm with a gripping force of 500 Newtons which can be useful in securing all cables with that range of motion. It's a self centering design which is useful for the grabbing of small ORU (mentioned in Interfaces subsection). Some few disadvantages is that it focuses mostly on the clamping capabilities, not much very functional, it requires a wrist and arm of proper size for flexibility in aligning

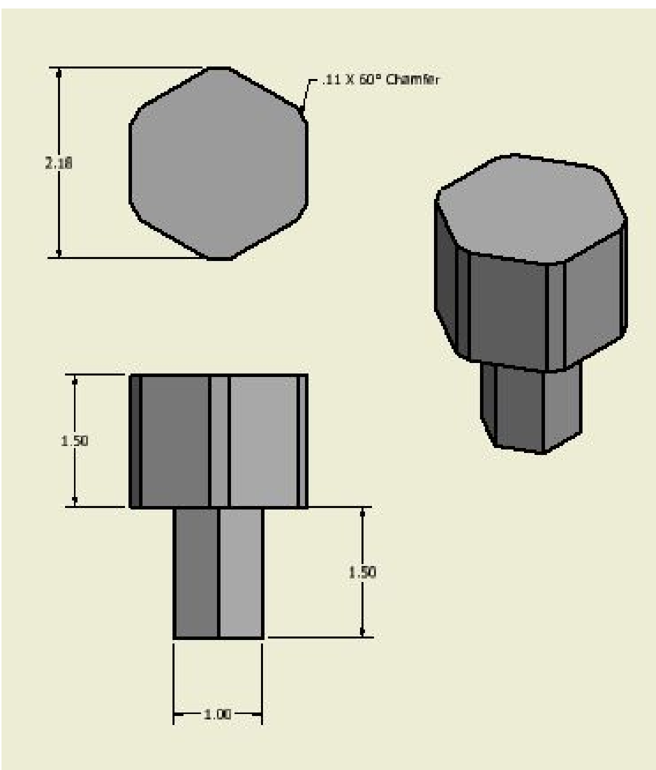


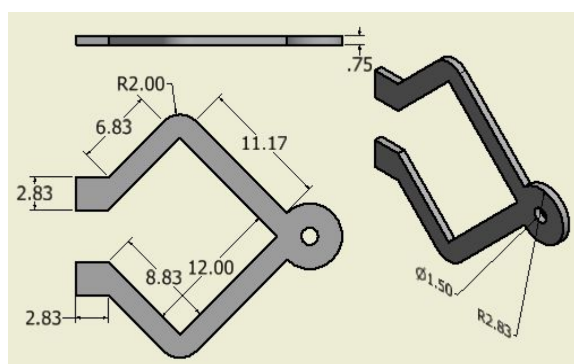
A 3-D printer will be considered for the end effector maintenance, as its main task is to fabricate the parts that may need to be replaced from the end effector. The printer will be taken into space, making the process of part replacement much faster and reliable.

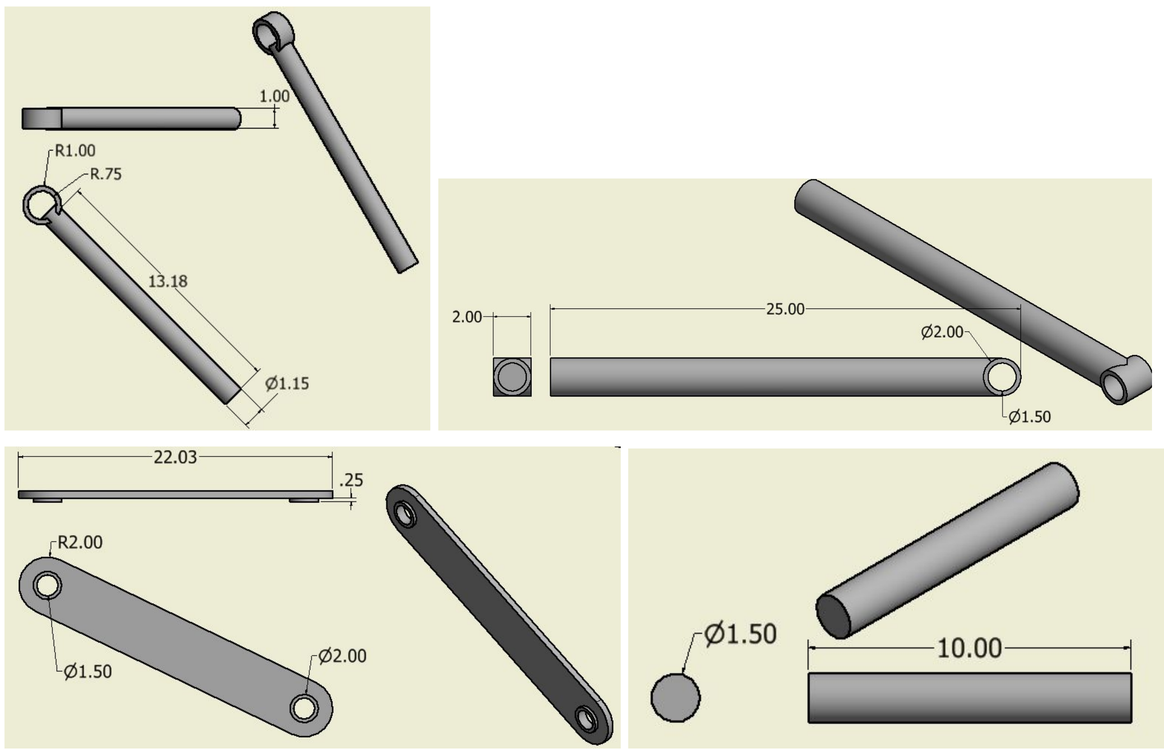
Tools: The tool we will use in TEXAS will be a 3D printed robotic-mechanical arm. The material the 3D printer will make the arm out of will be carbon nylon because it is durable, increases strength, lightweight, and prevents shrinking of parts after cooling ("Carbon Fiber Filled", 2019). The arm will be 3D printed to be lighter, cheaper, and will consist of hydraulic pistols. Hydraulics work with pressured fluids and therefore won't compress like gases up on the surface of Mars (Terranova, 2015). The fluid will have to be enclosed within the 3D printed arm before launching. Vegetable oil would be the best fluid for the arm to move because it is inexpensive, safe, and can function in vast temperature ranges (Green, 2018). Hydraulic pistols will be attached to arm, so it can move on the y-plane and at the end of the arm will be a rotating hex driver which will make the robot move on the z-plane (Green, 2018). The arm with the claw will provide better mobility to turn bolts or screws with a hexagonal shape, and therefore help with the reparations on a mission.





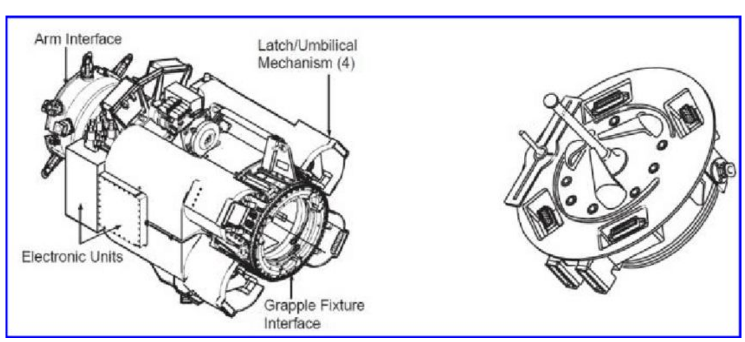






Interfaces: The Orbital Replacement Unit (ORU) is a key element to the ISS's success due to the fact that if an element were to fail, they would be easily replaceable either through an ExtraVehicular Activity (EVA) or by the Dextre and the CanadaArm2. ORU's are able to be manipulated by the Dextre and its Tool Changeout Mechanism, preventing the need for astronauts aboard the ISS to conduct EVA's. Currently, ORU's connect to the dextre through a variety of different means such as the Flight Releasable Grapple Fixture (FRGF), Latchable Grapple Fixture (LGF), Power and Video Grapple Fixture (PVGF), and the Power Data Grapple Fixture (PDGF).

The FRGF is the simplest grapple fixture that only allows for grappling and does not have any electrical connectors. The LGF allows for grappling and latching and does not have any electrical connectors. The PVGF allows for grappling and latching as well as electrical connectors for data, video, and power. The PDGF allows for grappling and latching as well as electrical connectors for data, video, and power. All 4 variants look relatively the same and are meant to be used with the Latching End-Effector. Both look as follows:



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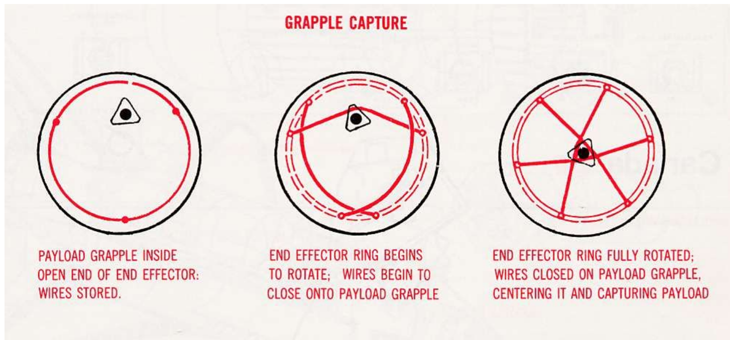
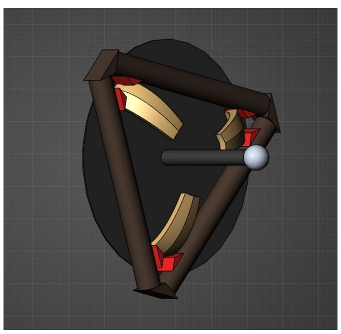
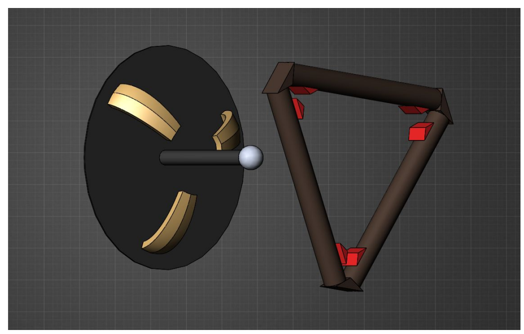
Because any tap in space can cause the object to move away indefinitely, standard pincer-like grippers would need to move with absolute precision and accuracy to avoid accidentally pushing it. For that the Latching End-Effector uses cables to align the rod from the fixture interface.



This system is simple and effective so no changes to the overall grappling system will be made; however, parts can be simplified. The latching system for example can be combined with alignment system and power system to save on weight.





In the figure above, brushes, shown in red, can be added to the alignment mechanism to transmit power, data, and video through the metal of the alignment mechanism. Latches can be added to the alignment mechanism similar to the clips on seat belts to fix the ORU to the LEE.

Abstract:

The TEXAS project incorporates an upgrade to the current Dextre arm through a reduction of weight and complexity while adding functionality with an improved tool exchange mechanism. The project timeline includes milestones at completion of major phases throughout the life of the project and include the modeling, prototyping, and testing phases. Both the modeling and prototyping phases have a 4 week duration, while the testing phase is 2 weeks, for a total of 10 weeks from initial work to completion.

Initial budgetary plans break the $10,000 lump sum into three categories: fabrication, testing, and travel. The fabrication and testing categories will receive the majority of the funding, owing to their innate importance to the project and allowing for material and labor costs. The remaining funds will go to funding transportation and lodging for visiting the Glenn Research Center in

Cleveland, Ohio. The initial funding will be broken down as follows: $4000 fabrication, $4000 testing, and $2000 travel. Each of the three categories can be further divided into smaller funding groups to effectively use resources.

Technical Merit and Work Plan:
The TEXAS extension and its accompanying tools is meant to be an addition to the current Dextre arm that is in service now. It's meant to deliver the same quality of work that Dextre arm currently provides while weighing less and reducing the complexity. The manipulators limitations is its force it can clamp down with, the maximum opening width, and that it can only handle square and cylindrical objects. The hydraulic claw will be able to rotate a full 360 degrees to the left and right, however because of the connected wires the claw is unable to rotate past 360 degrees. The FRGF is limited by its grip distance and that it is unable to send video or any data. The LGF can only grapple with no video while the PFGF can manipulate objects and latch while providing a video feed. The PDGF is the most complicated of the grappling fixtures and provides all the characteristics of the previous fixtures in one.

Our current technology limitations are the materials we can manufacture the tools out of, the quality of whatever 3D printer and its printing process, and the quality of the hydraulics and its fluid. The interface is limited by the bandwidth that the cables can carry, and the display its connected to.

If one of the designs were to change in the future the only parts that would need to be upgraded are the ones adjacent to the upgrade to ensure the lifetime of the upgraded part is maximized and any damage done during maintenance is negated. Because the parts are 3D printed they don't provide the same structural integrity. However they will be easily replaceable with the 3D printer on the ISS. As the 3D printed parts reach their end of life the damage and stress fractures will become visible over time. This provides easy assessment of when a part should be replaced and with routine inspection will give plenty of time to start a print of a part so a replacement will be ready before a damaged part fails.
Project Management Approach:

Milestones for the project include:

Model Completion Phase - 4 weeks
Prototyping Phase - 4 weeks
Testing Phase -2 weeks

During the Model Completion Phase, the team will be working on completing a manufacturable model and will be split into 3 sub-teams for the Manipulator, Tools, and Interfaces. Labor will consist of sit down modeling time. 4 weeks is necessary to complete proper models to account for any potential time lost due to work, school, and life. Skills required from team members are considerable knowledge of 3d modeling, assembly, and manufacturing methods. For example, knowing how to design a part to be milled vs a part to be laser cut.

During the Prototyping Phase, the team will choose one of the 3 models to be manufactured and where it will be manufactured. For example the Team may choose the manipulator model to be manufactured at UT Dallas in their fabrication laboratory because they have almost all the tools they have designed for with the exception of one that can be manufactured at UT Tyler and mailed to UT Dallas. Time required for this phase is 4 weeks because, in the event the machine is down for repairs or has a long queue until the part can be manufactured or training is needed to use the machine. Skills required include knowledge of machine use, often fabrication laboratories require that students themselves learn how to use the machine before they use it, drafting skills to send to on campus machinists, and shipping time for parts bought online.

During the Testing Phase, the team will test the prototype and make any adjustments as necessary. Time required is 2 weeks because testing only requires that the team uses the prototype and document its properties. In addition to the 2 weeks in-house testing time, part of our budget will be used to take the prototype and 2 team members to Glenn Research Center located in Cleveland, Ohio. The prototype will then go under a variety of tests such as but not limited to the Space Simulation Vacuum chamber, Electromagnetic Interference/Compatibility,

Teaming and Workforce Development:

Jason Cabrejos:
Role: Project Manager, interfaces member - purpose is to reduce the number of parts, weight, and/or cost while maintaining or improving current performance.
Experience: Arm and Effector Lead of the Rover Team at UTA. Jason Cabrejos has designed robotic arms for a student competition, meant to assist astronauts in object retrieval, equipment servicing, and proof of life. Time Jason has spent on the proposal includes many hours of research on ORU's, Tools changeout mechanism and issues on the ISS regarding the Dextre, ISS assembly, and ExtraVehicular Activities.
Jesse Sullivan:
Role: Logistics Manager - purpose, to evaluate and compare budgeting, as well as delegate roles to team members.
Experience: Sullivan was a CAD engineer on L'SPACE Academy 1 and 2 Team 20 lander group. Jesse Sullivan also has several years of experience in the competitive intercollegiate rocketry environment through the UTA Design-Build-Fly (DBF) student organization Aero Mavericks as vehicle design lead, project manager, and project mentor. Founder of UNT Eagle Aerospace student organization. Currently he has been researching costs for current programs he is taking place in.
Amer Khalaf:
Role: Research and document proposals alignment to NASA/National needs.
Experience: Khalaf has experience in technical writing such as explaining and instructing specific subject roles in projects. Khalaf, has spent and average of 42 hours working on the team project.
Miguel Gomez:

Role: Research implications needed for NASA proposal regarding NASA's current pathways and national needs.
Experience: Gomez has experience in programming and dealing with electronics. Also, he has participated in NASA research and project development. Gomez, has spent an average of 40 hours working on the team project.
Omar Asbun:
Role: Manipulator - reduce projects components weight while maintaining the functionality of the manipulator.
Experience: Omar has spent an average of 50 hours working on the team project.
Aaron Gonzalez:
Role: Deputy Project Manager - write technical merit and workplan for project proposal.
Experience: Gonzalez has past experience drawing and designing tools using CAD software. He has spent an average of 30 hours working on the team project.
Valeria Gonzalez Romero:
Role: Research a tool that will aid in reparations which will decrease cost, be a lighter tool, and be durable.
Experience: Gonzalez Romero is certified in AutoCAD, has successfully completed NASA Community College Aerospace Scholars phase one and two, and has created a mars rover on CAD software. Gonzalez has spent an average of 35 hours working on the team project.
Jorge Penaloza:
Role: Research how hydraulics tool will work with a 3D printer, what material to use, compiled the teams information into the teaming and workforce development.
Experience: Penaloza has experience working in CAD software and successfully completed phase one and two of NASA Community College Aerospace Scholars. Penaloza spent an average of 40 hours working on the team project.

Appendix:

Tool Effector eXchange Array System (TEXAS)

PI: Jason Cabrejos & Amer Khalaf, Team 11

Goal / Objective

- To minimize the need for emergency spacewalks by improving methods of detecting and dealing with potential failures and maintenance tasks.
- By standardizing construction methods and improving a specialized end effector designed to have a diverse toolkit needed to make efficient and long-lasting repairs.
- Improved version of Dextre's manipulators, tools, and interfaces.

Team Overview

- We have Mechanical, Electrical, Aerospace, and Computer Engineering and Science majors on our team. Some members have also been/currently are involved in robotics.
- 7.2.2 Maintenance System: Design and optimize specialized processes and tools that are required to perform maintenance during human exploration missions.

Metrics and Key Performance Parameters

- The improved Dextre will reduce the number of parts, reduce cost, and make it easier to use for future astronauts on the gateway and other bases.
- The improvements will have a positive impact on NASA schedules by reducing EVA's to allow more time for onboard science.
- Current Dextre Specs; Cost: $200M, DOF: 15, Mass: 1710 Kg, Power to keep alive: 600 W

“Carbon Fiber Filled.” *All-In-One 3D Printing Software*, Simplify3D Software, 30 May 2019, www.simplify3d.com/support/materials-guide/carbon-fiber-filled/.

Green, J. (2018, July 11). *Testing Validates Environmentally Friendly Hydraulic Fluids*. Retrieved from https://www.hydraulicspneumatics.com/hydraulic-fluids/testing-validates-environmentally-friendly-hydraulic-fluids

Terranova, A. (2015, October 26). *Hydraulic Robots*. Retrieved from https://makezine.com/2015/05/16/skill-builder-hydraulics-robots/

“Robot Arm's Payload Acquiring Sequence.” *Robot Arm's Payload Acquiring Sequence*, JAXA, 28 Mar. 2000, http://iss.jaxa.jp/iss/3a/orb_rms2_e.html.

National Aeronautics and Space Administration	# Disclosure of Invention and New Technology (Including Software)	**Form Approved O.M.B. NO. 2700-0009**	DATE **10/22/2019**
		CONTRACTOR CASE NO.	

This is an important legal document. Carefully complete and forward to the Patent Representative (NASA in-house innovation) or New Technology Representative (contractor/grantee innovation) at NASA. Use of this report form by contractor/grantee is optional; however, an alternative format must at a minimum contain the information required herein. NASA in-house disclosures should be read, understood and signed by a technically competent witness in the witness signature block at the end of this form. In completing each section, use whatever detail deemed appropriate for a "full and complete disclosure." Contractors/Grantees please refer to the New Technology or Patent Rights – Retention by the Contractor clauses. When necessary, attach additional documentation to provide a full, detailed description.

NASA CASE NO. (OFFICIAL USE ONLY)

1. DESCRIPTIVE TITLE
Tool Effector eXchange Array System (TEXAS)

2. INNOVATOR(S) *(For each innovator provide: Name, Title, Work Address, Work Phone Number, and Work E-mail Address. If multiple innovators, number each to match Box 5.)*
#1 Jason Cabrejos, Project Manager, 604 Parkway Blvd, Coppell TX, 702-810-1897, jasoncabrejos@hotmail.com
#2 Amer Khalaf, Alignment to Agency, 2811 Hayes Rd, Houston, TX 77082, 832-846-9922, aahk360@me.com
#3 Aaron Gonzalez, Technical Merit, 152 Serrato Cove Drirftwood, TX 78619, 832-574-7693, ajg248@txstate.edu
#4 Jesse Sullivan, Abstract, 14796 Pensham Drive, Frisco, TX, 469-422-8425, jessesullivan@my.unt.edu
#5 Jorge Penaloza, Tools and Workforce Development, 12010 Barrycliff Ct, Houston, TX, 77070, 713-325-1608, penal.jorge@gmail.com
#6 Valeria Gonzalez, Tools and Appendix, 2207 Louetta Falls Ln, Spring, TX, 77388, 832-916-9819, valerie5471@sbcglobal.net
#7 Miguel Gomez, National Needs, 14051 Imperial Canyon Ln, Sugar Land, TX, 77498, 281-818-6861, magomezg95@gmail.com
#8 Omar Asbun, Manipulator, 1312 Bennett Dr, Arlington, TX, 76013, 682-701-9926, omarasbun95@gmail.com

3. INNOVATOR'S EMPLOYER WHEN INNOVATION WAS MADE *(For each innovator provide: Name, Division and Address of Employer, Organizational Code/Mail Code, and Contract/Grant Number if applicable. If multiple innovators, number each to match Box 5.)*
L'Space, 604 Parkway Blvd, Coppell TX, 80MSFC18N0002

4. PLACE OF PERFORMANCE *(Address(es) where innovation made)*
604 Parkway Blvd, Coppell TX

5. EMPLOYER STATUS *(choose one for each innovator)*

Innovator #1	Innovator #2
CU	CU
Innovator #3	**Innovator #4**
CU	CU

GE = Government
CU = College or University
NP = Non-Profit Organization
SB = Small Business Firm
LE = Large Entity

6. ORIGIN *(Check all that apply and provide all applicable numbers. If multiple Contracts/Grants, etc., list Contract/Grant Numbers in Box 3 with applicable employer information.)*
☐ NASA In-house Org. Mail Code ____ WBS ____
WBS ____
☒ Grant/Cooperative Agreement No. 80MSFC18N0002 WBS ____
☐ Prime Contract No. ____
Task No. ____ Report No. ____ WBS ____
☐ Subcontractor; Subcontract Tier ____
☐ Joint Effort *(contractor, subcontractor and/or grantee contribution(s), and NASA in-house contribution)*
☐ Multiple Effort *(multiple contractor, subcontractor and/or grantee contributions, no NASA in-house contribution)* WBS ____
☐ Other (e.g., Space Act Agreement, MOA) No. ____

7. NASA CONTRACTING OFFICER'S TECHNICAL REPRESENTATIVE *(COTR)*
N/A

8. CONTRACTOR/GRANTEE NEW TECHNOLOGY REPRESENTATIVE *(POC)*

9. BRIEF ABSTRACT *(A general description of the innovation which describes its capabilities, but does not reveal details that would enable duplication or imitation of the innovation.)*

The TEXAS project incorporates an upgrade to the current Dextre arm through a reduction of weight and complexity while adding functionality with an improved tool exchange mechanism. The project timeline includes milestones at completion of major phases throughout the life of the project and include the modeling, prototyping, and testing phases. Both the modeling and prototyping phases have a 4 week duration, while the testing phase is 2 weeks, for a total of 10 weeks from initial work to completion.

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SECTION I – DESCRIPTION OF THE PROBLEM OR OBJECTIVE THAT MOTIVATED THE INNOVATION'S DEVELOPMENT ***(Enter as appropriate: A. – General description of problem/objective; B. – Key or unique problem characteristics; C. – Prior art, i.e., prior techniques, methods, materials, or devices performing function of the innovation, or previous means for performing function of software; and D. – Disadvantages or limitation of prior art.)***

SECTION II – TECHNICALLY COMPLETE AND EASILY UNDERSTANDABLE DESCRIPTION OF INNOVATION DEVELOPED TO SOLVE THE PROBLEM OR MEET THE OBJECTIVE ***(Enter as appropriate; existing reports, if available, may form a part of the disclosure, and reference thereto can be made to complete this description: A. – Purpose and description of innovation/software; B. – Identification of component parts or steps, and explanation of mode of operation of innovation/software preferably referring to drawings, sketches, photographs, graphs, flow charts, and/or parts or ingredient lists illustrating the components; C. – Functional operation; D. – Alternate embodiments of the innovation/software; E. – Supportive theory; F. – Engineering specifications;***
G. – Peripheral equipment; and H. – Maintenance, reliability, safety factors.)

SECTION III – UNIQUE OR NOVEL FEATURES OF THE INNOVATION AND THE RESULTS OR BENEFITS OF ITS APPLICATION ***(Enter as appropriate: A. – Novel or unique features; B. – Advantages of innovation/software; C. – Development or new conceptual problems; D. – Test data and source of error; E. – Analysis of capabilities; and F. – For software, any re-use or re-engineering of existing code, use of shareware, or use of code owned by a non-federal entity.)***

SECTION IV – SPECULATION REGARDING POTENTIAL COMMERCIAL APPLICATIONS AND POINTS OF CONTACT (*Including names of companies producing or using similar products.)*

10. ADDITIONAL DOCUMENTATION *(Include copies or list below any pertinent documentation which aids in the understanding or application of the innovation (e.g., articles, contractor reports, engineering specs, assembly/manufacturing drawings, parts or ingredients list, operating manuals, test data, assembly/manufacturing procedures, etc.).)*

TITLE	PAGE	DATE

11. DEGREE OF TECHNOLOGY SIGNIFICANCE (*Which best expresses the degree of technological significance of this innovation?)*

☒Modification to Existing Technology ☐ Substantial Advancement in the Art ☐ Major Breakthrough

12. STATE OF DEVELOPMENT

☐ Concept Only ☐Design ☐Prototype ☐Modification ☐Production Model ☐Used in Current Work

13. PATENT STATUS (Prior patent on/or related to this innovation.)

☐Application Filed Application No. _____ Application Date _____

☐Patent Issued Patent No. _____ Issue Date _____

14. INDICATE THE DATE OR THE APPROXIMATE TIME PERIOD WHICH THIS INNOVATION WAS DEVELOPED (*i.e., conceived, constructed, tested, etc.)*

15. PREVIOUS OR CONTEMPLATED PUBLICATION OR PUBLIC DISCLOSURE INCLUDING DATES *(Provide as applicable: A. – Type of publication or disclosure, e.g., report, conference or seminar, oral presentation; B. – Disclosure by NASA or Contractor/Grantee; and C. – Title, volume no., page no., and date of publication.)*

16. QUESTIONS FOR SOFTWARE ONLY

(a) Using non-NASA employees to beta-test the program? ☐YES ☐NO If Yes, done under a beta-test agreement? ☐YES ☐NO

(b) Modification of this program continued by civil servant and/or contractual agreement? ☐YES ☐NO

(c) Copyright registered? ☐YES ☐NO ☐UNKNOWN If Yes, then by whom? _____

(d) Has the latest version been distributed outside of NASA or contractor? ☐YES ☐NO ☐ UNKNOWN
If Yes, date of first disclosure: _____

(e) Were prior versions distributed outside of NASA or Contractor? ☐YES ☐NO If Yes, supply NASA or contractor contract: _____

(f) Contains or based on code not owned by U.S. Government or its contractors? ☐YES ☐NO ☐ UNKNOWN
If Yes, name of code and code's owner: _____

Has a license for use been obtained? ☐YES ☐NO ☐ UNKNOWN

17. DEVELOPMENT HISTORY

STAGE OF DEVELOPMENT	DATE (MM/YYYY)	LOCATION	IDENTIFY SUPPORTING WITNESSES (NASA in-house only)
a. First disclosure to others			
b. First sketch, drawing, logic chart or code			
c. First written description			
d. Completion of first model of full size device *(invention)* or beta version *(software)*			

e. First successful operational test *(invention)* or alpha version *(software)*			

f. Contribution of innovators *(if jointly developed, provide the contribution of each innovator)*

g. Indicate any past, present, or contemplated government use of the innovation

18. SIGNATURES OF INNOVATOR(S), WITNESS(ES), AND NASA APPROVAL

TYPED NAME AND SIGNATURE *(Innovator #1)*		DATE	TYPED NAME AND SIGNATURE *(Innovator #2)*	DATE
TYPED NAME AND SIGNATURE *(Innovator #3)*		DATE	TYPED NAME AND SIGNATURE *(Innovator #4)*	DATE
TYPED NAME AND SIGNATURE *(Witness #1)*		DATE	TYPED NAME AND SIGNATURE *(Witness #2)*	DATE
NASA APPROVED	TYPED NAME		SIGNATURE	DATE